Retail Trading on Post Shock Price Reversion

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I. Abstract

This study examines the relationship between retail trading activity in conjunction with security price shocks and their five-day post-shock price behavior. The hypothesis is that price shocks driven by uninformed retail trading may be prone to overreactions and subsequent reversals in price. The hypothesis suggests that increases in retail trading on days observed with abnormal price shocks will cause an increase in price reversals in the subsequent five day period. Using data from Wharton Research Centers MSEC and CRSP databases, I have implemented a date fixed-effects regression analyzing a sample of approximately 185 stocks over the period of January 2019 to 2024, and using Oddlot rates as a proxy for retail investor activity. In all methods examined, retail trading was associated with negative cumulative abnormal returns in the five days following observed price shocks. This relationship is asymmetric and finds retail participation to have stronger and more significant effects in cases of negative initial price moves. In cases of positive price shocks retail investment is also associated with lower five-day CAR but this effect loses significance as shocks become more extreme. According to the theory of retail induced price reversion, we would have expected to see positive CAR rather than negative in negative price shocks. Therefore, I have found the underlying hypothesis to be null.

II. Introduction

In the last decade, investment platforms have become increasingly accessible to individual retail investors. Retail Investors are “investors who buy and sell securities for personal accounts rather than institutional or business purposes” (*Retail Investors*, 2024). The number of individuals moving money into investment accounts has increased by several times since the mid 2010’s. (Wheat, C. & Eckerd, G., 2025). This rapid growth can be attributed to several key factors regarding account accessibility. Over the last decade, investment minimums, trading costs, and commissions have fallen with platforms such as Robinhood offering zero commission trading upon its launch in 2013 and introducing split share trading as of December 2019. Split share trading allows individuals to invest in high value stocks with as little as one dollar. These changes, pitched by Robinhood as the democratization of financial markets, have allowed people to invest without significant financial barriers to entry. The concept often raises concerns about higher risks for novice investors operating without fundamental knowledge of balancing risks and returns in the market (Greene, 2021).

There are two distinct schools of thought around the impact of retail traders on the market. The first is grounded in the Efficient Market Hypothesis which suggests prices instantaneously reflect all available information, so retail trading does not systematically affect valuation (Fama, E.F., 1970). The latter is the Noise Trader Theory, which argues that not all investors behave rationally and arbitrage by those that do is “risky and limited.” Therefore, the correcting effect of rational investors on the market may not be captured instantaneously (J Bradford De Long et al., 1990).

Previous studies suggest that investors trade in predictable and irrational ways in response to non-fundamental news (Heston, S. L. & Sinha N.R., 2016). Furthermore, literature examining market inefficiencies demonstrates that non-fundamental noise trading can actively impede efficient price discovery in highly liquid securities like the SPDR S&P 500 (Berton, S., 2011), and in Real Estate Investment Trusts (REITs) (Yung, K. & Nafar, N., 2017). Another study found that large price increases (decreases) in conjunction with high (low) abnormal trading volumes are associated with significant subsequent price reversals (Kudryavtsev, A., 2019).

The specific role of the retail trading proportion of volume remains an area for further investigation; this paper contributes to the literature by isolating the retail component. We examine oddlots rates as a proxy for individual investor activity to understand its effect on price reversals. While previous studies focus on either long-term speculative bubbles or frequent intraday fluctuations; this study examines a five-day horizon. This timeframe was selected to capture the window in which transient price pressure, driven by retail sentiment, is absorbed by institutional arbitrageurs, as they enter the market. The hypothesis is: Increases in retail participation in conjunction with abnormal price shocks will be associated with significant price reversals in the subsequent five trading days.

III. Theory

The Efficient Market Theory suggests: When traders deviate from rational behavior and transact based upon sentiment or cognitive bias, rational traders should observe opportunities for arbitrage and react accordingly. This reaction through the form of security transactions or options trading should instantaneously adjust the price of the security back to its fundamental valuation.

In reality these market adjustments do not always happen instantaneously as extreme momentum driven conditions can cause high volatility, leading to risk-averse behavior by institutional arbitrageurs (Shleifer & Vishny, 1997). In addition, we see greater risk for arbitraging investors as sentiments and cognitive bias by nature may be difficult to predict, therefore doubts over the persistence of price momentum constrain arbitrage through uncertainty over future prices (Shleifer & Summers, 1990).

This paper examines how retail trading participation rates affect five-day price behavior following abnormal price shocks. It uses Oddlot Rates as a proxy for retail participation under the assumption that oddlot trades are predominantly executed by individual investors. The theory is that shocks with a greater composition of retail trading volume on that day may be representative of uninformed sentiment as retail traders are reacting to events in greater proportion than institutional investors.

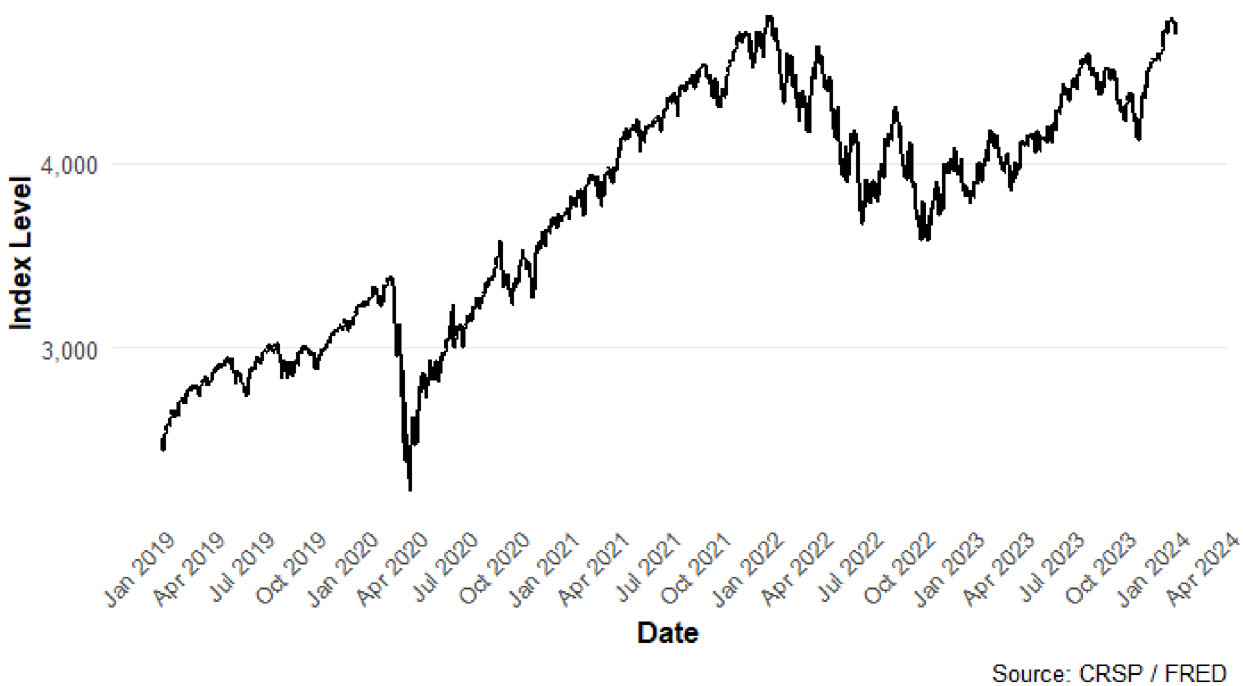
My theory is price shocks driven by uninformed sentiment are more likely to be corrected by efficient market forces as institutional investors enter the market. These sentiment-based reactions are expected to occur asymmetrically as positive shocks may be amplified by attention driven buying, (Barber & Odean, 2008) while negative shocks may be influenced by the loss-realization aversion aspect of the disposition effect. (Shefrin & Statman, 1985) The underlying hypothesis is that increases in retail trading around price shocks will increase the magnitude of price reversals as the initial shocks are driven by transient price pressure rather than fundamental price discovery.

IV. Data

The data used in this paper was sourced from the Wharton Research Data Services (WRDS) MSEC and CRSP databases. The final dataset consists of daily price, volume, and odd-lot data for 185 NYSE-listed securities from January 1, 2019, to January 5, 2024. The selected sample of securities includes a diversified mix of market capitalizations and sectors (see

Appendix A). Additionally, daily S&P 500 prices were collected from the FRED database to serve as a proxy for market performance (Figure 1).

Figure 1: S&P 500 Performance



Several key variables were constructed for empirical analysis:

- **Idiosyncratic_Ret :** The simple market-adjusted return calculated as the daily percent difference between the individual securities return and the S&P 500 return.
- **CAR [1,5] :** The cumulative average percent idiosyncratic returns over the five trading days following the observation.
- **Retail_Intensity :** Calculates the number of standard deviations the daily Oddlot Rate deviates from its full sample mean to create a standardized measure of retail activity.

- **Shock_Intensity :** Categorical variable that assigns price shocks into bins based upon their direction and magnitude. This allows the examination of non-linear effects around extreme price movements. The categorical bins and their parameters appear in Table 1.

Table 1:

Shock Type	**Parameter**	**Observation Count**
Extreme Positive	(z > 2)	5115
Moderate Positive	(z > 1)	17900
Positive	(z > 0)	83263
Negative	(z < 0)	88291
Moderate Negative	(z < -1)	18367
Extreme Negative	(z < -2)	4335

V. Methodology

The empirical analysis utilizes a multi-linear fixed-effect regression model to examine the relationship between retail trading concentration and subsequent price reversals. The model is designed as follows:

$$CAR[1,5] = \beta_0 + \boldsymbol{\beta}\ (Retail\ Intensity_{i,t} \times Shock\ Intensityy_{i,t}) + \gamma X_{i,t} + \alpha$$

- The **CAR[1,5]** variable captures the five-day cumulative idiosyncratic return following the observed day.
- **Retail Intensity x Shock Intensity** The set of interaction variables representing standardized retail activity and the six categorical magnitudes/directions of the initial

shock. This distinction allows the observation of asymmetric behavior across different extremes.

- $\gamma X_{i,t}$ is a vector of control variables including log transformed trade volume log(VOL) and share price log(PRC), and historical volatility (sd_pct_change). These controls isolate the compositional effect of retail traders from general liquidity and price-level biases as well as handling the differences between stocks of different volatilities.
- α represents the date fixed effects which are used to handle time-varying factors like market-wide shocks that affect all securities in the sample simultaneously. This is necessary as the period of study includes intense market fluctuations especially during COVID-19 which could otherwise skew the results.

The use of the interaction model is critical for testing the Noise Trader Hypothesis, as it allows for the observation of asymmetric market reactions. Specifically, it enables us to determine if the predictive power of retail sentiment regarding price reversals varies between positive and negative shocks, or between moderate and extreme price events. The model clusters standard errors by tickers to account for correlations within firms. The result of this ensures that the significance of retail trading is not inflated by intra-firm patterns.

VI. Results and Analysis

Table 2 presents the results of the fixed-effects regression model analyzing the impact of retail trading intensity on five-day Cumulative Abnormal Returns (CAR [1,5]) following idiosyncratic price shocks. This model includes date fixed effects and standard errors clustered at the ticker level to account for market-wide shocks and intra-firm correlations.

Table 2:

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001		
Coefficients	**Estimate**	**Std Error**
retail_intensity	-0.002*	-0.001
Shock_IntensityNegative	0.005***	-0.001
Shock_IntensityModerate Negative	0.006***	-0.001
Shock_IntensityExtreme Negative	0.005*	-0.002
Shock_IntensityPositive	0.004**	-0.001
Shock_IntensityModerate Positive	0.005***	-0.001
log(VOL)	0	0
log(PRC)	0	0
sd_pct_change	0.134***	-0.029
retail_intensity × Shock_IntensityNegative	-0.003**	-0.001
retail_intensity × Shock_IntensityModerate Negative	-0.002**	-0.001
retail_intensity × Shock_IntensityExtreme Negative	-0.006***	-0.002
retail_intensity × Shock_IntensityPositive	-0.002*	-0.001
retail_intensity × Shock_IntensityModerate Positive	-0.002+	-0.001
retail_intensity × Shock_IntensityExtreme Positive	-0.004	-0.003
Num.Obs.	198600	
R2	0.035	
R2 Adj.	0.029	
R2 Within	0.003	
R2 Within Adj.	0.003	
RMSE	0.07	
Std.Errors	by: TICKER	

The baseline coefficient for retail_intensity is negative and significant (β = -0.002, p < 0.05). This suggests increases in retail participation is associated with slight downward pressure on subsequent five-day CAR when held independent of specific price shock directions and magnitudes.

The primary finding of the study is negative returns are significantly amplified when retail intensity coincides with negative price shocks. In cases of Extreme Negative shocks the effect of retail intensity is significant ($\beta = -0.006$, $p < 0.001$) and because the initial shock is negative the negative interaction estimate indicates a momentum effect rather than reversal. Across all negative shock categories high retail trading concentration is attributed with lowered performance in the five-day CAR.

This finding suggests retail investors may drag down prices during downturns.

In contrast, the results around positive price shocks provide evidence of mean reversion. The interactions for Positive and Moderate Positive shocks are negative and significant ($p < 0.05$, $p < 0.1$) and because the initial shock was positive these negative coefficients suggest retail intensity is associated with post shock price reversion over the following days. During Extreme Positive events this effect loses significance. This result could represent an asymmetric change in market behavior caused by expectations about future pricing or simply be an effect of the model not having enough extreme cases to examine.

The model controls for Stock Volatility (sd_pct_change) and log transformed Volume and Price. Stock Volatility was a highly significant predictor of future returns ($p < 0.001$), while log transformed Volume and Price were associated with near-zero coefficients. Their inclusion was important because it ensures the observed effect of retail trading was not only a result of greater liquidity or price-level differences.

VII. Conclusion, Limitations, and Future Research

The study demonstrates that days of increased retail trading intensity, proxied by oddlot rates, are associated with lower cumulative abnormal returns over the subsequent five trading days (CAR[1,5]). This inverse relationship persists across days with idiosyncratic returns of all directions and magnitudes. The effect is most pronounced in periods following strong negative price shocks.

While initially, I theorized that retail participation, in conjunction with price shocks, would cause price reversals, the data around negative shocks suggests the opposite. In fact the results find retail activity associated with greater price momentum following negative price movements. However, in cases of positive shocks the results do support a reversal effect as retail intensity is associated with a CAR[1,5] value in the opposite direction (negative).

Together the results suggest that retail traders may be driven by a "disposition effect" or lack of holding persistence. Retail Traders are known to sell early to realize quick gains in cases of upward price movement or to mitigate losses during downturns. Both scenarios result in sell-side pressure. This excess of supply relative to demand may be the cause of the short term downward price pressure associated with Retail Trading.

The results of the study are subject to several limitations which could be improved upon in future research:

- **Proxy Validity:** The primary limitation lies in the use of odd-lot rates as a proxy for retail activity. Historically this was a reliable method but the growth of high frequency algorithmic trading has led to greater use of oddlot executions by institutional investors. This introduces noise into the retail intensity metric making it less accurately represent the activity of individual investors.

- **Temporal Constraints:** The observed five-day window (CAR[1,5]) may fail to capture the behavior of retail-induced price pressure. Sentiment-driven movements can be reflected in prices over differing periods of time and the study fails to examine intraday movements or long term shifts.
- **Behavioral Assumptions:** The model assumes that lower-volume traders are inherently less informed. This assumption predicts irrational behavior associated with retail trading activity. Sophisticated "sub-institutional" investors may challenge this assumption.

These limitations could be addressed in future studies by implementing high-frequency intraday data with a trade size filter (e.g. observing transactions under $5,000) to more accurately represent individual retail activity. Furthermore, expanding the time horizon to include both intraday and long-term intervals would provide a more comprehensive understanding of how retail sentiment integrates into market pricing over time.

VIII. References

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Appendix A

Stock Sample: January 1st, 2019 to January 5th, 2024

A.	TICKER	Mean_Price	Mean_Oddlot_Rate	SD_Percent_Change	Obs_Count
1	AA	33.61	50.92	0.04	1261
2	AAON	62.44	81.84	0.02	1261
3	AAPL	187.42	50.90	0.02	1261
4	ABT	102.89	61.57	0.02	1261
5	ADBE	424.10	83.43	0.02	1261
6	ADI	146.33	74.32	0.02	1261
7	AIT	92.91	84.25	0.02	1261
8	AJG	143.32	79.34	0.02	1261
9	AKAM	97.11	74.81	0.02	1261
10	ALL	114.18	74.09	0.02	1261
11	AMC	12.68	34.89	0.12	1261
12	AMD	78.30	43.93	0.03	1261
13	AMZN	1843.18	81.41	0.02	1261
14	ARNC	24.59	NaN	0.03	313
15	BA	233.01	72.44	0.03	1261
16	BAC	32.71	25.46	0.02	1261
17	BB	6.68	NaN	0.04	1261
18	BBY	86.91	69.50	0.02	1261
19	BDX	249.55	80.55	0.02	1261
20	BIDU	142.86	NaN	0.03	1261
21	BIIB	267.84	82.63	0.03	1260
22	BIO	471.76	91.73	0.02	1261
23	BKNG	2157.53	94.90	0.02	1261

24	BLK	647.27	89.88	0.02	1261
25	BYND	80.30	67.95	0.05	1178
26	CAT	187.84	73.01	0.02	1261
27	CB	171.86	NaN	0.02	1261
28	CDNS	138.48	77.03	0.02	1261
29	CL	75.39	56.40	0.01	1261
30	CLOV	4.04	24.87	0.06	753
31	CLX	165.15	79.27	0.02	1261
32	CMI	208.32	81.03	0.02	1261
33	COIN	137.32	76.28	0.06	687
34	COOP	31.51	NaN	0.03	1261
35	COST	412.21	84.08	0.01	1261
36	CRM	197.03	71.93	0.02	1261
37	CRS	40.38	77.59	0.03	1261
38	CRWD	153.67	73.88	0.04	1150
39	CSCO	49.25	37.64	0.02	1261
40	CSX	54.67	46.12	0.02	1261
41	CVX	126.22	62.52	0.02	1261
42	DAL	41.39	45.92	0.03	1261
43	DE	294.22	78.59	0.02	1261
44	DEAC	11.76	NaN	0.05	201
45	DIS	127.91	64.34	0.02	1261
46	DKNG	32.80	48.90	0.05	932
47	DOCS	41.36	69.02	0.05	637
48	DOCU	117.71	71.26	0.04	1261

49	EBAY	48.28	48.97	0.02	1261
50	EMR	80.99	64.99	0.02	1261
51	ENPH	129.97	69.10	0.05	1261
52	ETN	134.28	NaN	0.02	1261
53	ETSY	114.26	71.52	0.04	1261
54	FB	243.80	66.72	0.03	865
55	FDX	212.59	77.50	0.02	1261
56	FI	122.86	NaN	0.01	147
57	FIS	109.04	64.73	0.02	1261
58	FISV	103.98	66.95	0.02	1114
59	FIVE	154.64	78.93	0.03	1261
60	FIZZ	53.95	81.07	0.03	1261
61	FNJN	2.09	NaN	0.04	392
62	FSLR	98.42	72.47	0.03	1261
63	FTNT	137.82	74.42	0.03	1261
64	FUBO	12.39	34.08	0.07	815
65	GIS	63.88	57.01	0.01	1261
66	GME	56.80	55.10	0.09	1261
67	GOOGL	1331.31	83.02	0.02	1261
68	GWRE	96.24	78.38	0.02	1261
69	HD	280.42	76.61	0.02	1261
70	HLT	118.74	70.84	0.02	1261
71	HON	187.24	74.52	0.02	1261
72	HOOD	14.99	37.94	0.05	613
73	HQY	64.69	74.88	0.03	1261

74	HSY	178.55	80.42	0.01	1261
75	HUM	410.67	84.59	0.02	1261
76	HWM	33.42	NaN	0.03	948
77	IBM	133.71	68.20	0.02	1261
78	INSM	25.20	65.90	0.04	1261
79	INTC	46.94	35.91	0.02	1261
80	INTL	41.44	NaN	0.03	378
81	IPOA	10.32	NaN	0.01	206
82	IPOC	11.19	NaN	0.03	144
83	ISDR	19.34	NaN	0.03	1261
84	ISRG	487.52	88.65	0.02	1261
85	ITW	201.16	80.28	0.02	1261
86	JD	53.24	NaN	0.03	1261
87	JLL	165.82	85.12	0.03	1261
88	JNJ	155.82	64.03	0.01	1261
89	JPM	129.74	58.00	0.02	1261
90	KLAC	293.49	82.68	0.03	1261
91	KMB	133.09	76.19	0.01	1261
92	KO	55.29	42.34	0.01	1261
93	KOSS	6.61	62.18	0.15	1261
94	L	52.80	72.02	0.02	1261
95	LMT	394.77	85.22	0.02	1261
96	LOW	171.50	71.63	0.02	1261
97	LRCX	445.35	84.39	0.03	1261
98	LULU	308.37	81.51	0.03	1261

99	LUV	43.32	50.68	0.03	1261
100	LW	77.50	70.71	0.02	1261
101	LYFT	33.84	47.10	0.04	1201
102	MA	330.25	78.16	0.02	1261
103	MAR	144.90	74.97	0.03	1261
104	MCD	233.43	76.47	0.01	1261
105	MDT	100.06	NaN	0.02	1261
106	MET	55.34	55.34	0.02	1261
107	META	164.24	71.19	0.03	543
108	METV	9.41	NaN	0.02	486
109	MKC	115.36	77.43	0.02	1261
110	MLI	46.18	77.82	0.03	1261
111	MMM	153.82	74.34	0.02	1261
112	MORN	208.47	90.39	0.02	1261
113	MOS	33.40	50.82	0.03	1261
114	MRK	87.78	52.76	0.01	1261
115	MRVL	44.14	61.04	0.03	1261
116	MSFT	236.79	63.53	0.02	1261
117	MTD	1117.42	94.72	0.02	1261
118	MTSI	49.07	74.20	0.03	1261
119	MU	61.56	47.47	0.03	1261
120	MWA	12.20	47.28	0.02	1261
121	NFLX	402.24	81.37	0.03	1261
122	NKE	114.58	64.32	0.02	1261
123	NOC	384.20	84.89	0.02	1261

124	NOK	4.66	NaN	0.03	1261
125	NOW	449.24	83.33	0.03	1261
126	NSC	222.99	79.43	0.02	1261
127	NVDA	313.36	74.93	0.03	1261
128	ORCL	74.16	47.72	0.02	1261
129	PANW	302.74	80.45	0.02	1261
130	PCAR	82.31	70.96	0.02	1261
131	PEP	152.75	69.79	0.01	1261
132	PFE	41.04	32.48	0.02	1261
133	PLTR	15.98	31.52	0.05	821
134	PLUG	15.75	33.88	0.05	1261
135	PRU	91.86	73.76	0.02	1261
136	PTON	46.93	50.10	0.05	1076
137	QCOM	115.80	65.30	0.03	1261
138	RBLX	53.04	61.16	0.05	711
139	REGN	576.90	89.23	0.02	1261
140	RGA	128.82	82.73	0.03	1261
141	RIOT	12.20	35.80	0.07	1261
142	RL	107.17	76.54	0.03	1261
143	RSG	112.75	76.02	0.01	1261
144	RTX	83.70	61.74	0.02	946
145	SBUX	92.97	60.30	0.02	1261
146	SEDG	203.32	78.87	0.04	1261
147	SFM	26.67	55.39	0.02	1261
148	SHOP	562.50	NaN	0.04	1261

149	SNA	200.50	85.39	0.02	1261
150	SNAP	25.99	32.07	0.05	1261
151	SNDL	1.58	NaN	0.08	1115
152	SNEX	75.50	89.29	0.02	883
153	SNOW	215.28	78.45	0.04	831
154	SNPS	269.16	84.11	0.02	1261
155	SPCE	14.31	38.17	0.06	1055
156	SPGI	338.24	82.54	0.02	1261
157	SQ	114.84	63.92	0.04	1261
158	SSB	73.73	81.82	0.03	1261
159	SYK	233.81	80.02	0.02	1261
160	T	25.96	28.10	0.02	1261
161	TEX	37.39	68.62	0.03	1261
162	TJX	65.73	54.05	0.02	1261
163	TMO	455.70	83.61	0.02	1261
164	TROW	132.94	78.17	0.02	1261
165	TSLA	538.34	80.30	0.04	1261
166	TSN	70.80	67.30	0.02	1261
167	TWLO	168.45	73.50	0.04	1261
168	UAL	53.22	53.35	0.04	1261
169	UBER	38.42	39.26	0.04	1172
170	ULTA	349.67	85.00	0.03	1261
171	UNH	393.09	78.48	0.02	1261
172	UPS	158.36	72.86	0.02	1261
173	UTX	132.74	NaN	0.02	315

174	V	205.08	71.42	0.02	1261
175	VRSN	203.20	85.47	0.02	1261
176	VRTX	251.33	81.06	0.02	1261
177	VZ	50.77	36.38	0.01	1261
178	WAT	274.27	86.15	0.02	1261
179	WM	136.52	76.87	0.01	1261
180	WMT	134.26	62.56	0.01	1261
181	WORK	32.14	35.50	0.04	524
182	XOM	75.30	42.90	0.02	1261
183	YUM	113.34	73.12	0.02	1261
184	ZBRA	321.02	87.77	0.03	1261
185	ZM	171.87	74.83	0.04	1187
186	ZS	146.64	74.12	0.04	1261